Communicating with Data – Introduction to Digital & Interactive Visualization

Dr. Ab Mosca (they/them)

Plan for Today

- Digital Visualization
  - Examples
  - Compare & contrast with print
- Interaction
  - Definition
  - Taxonomies of interaction
  - Examples

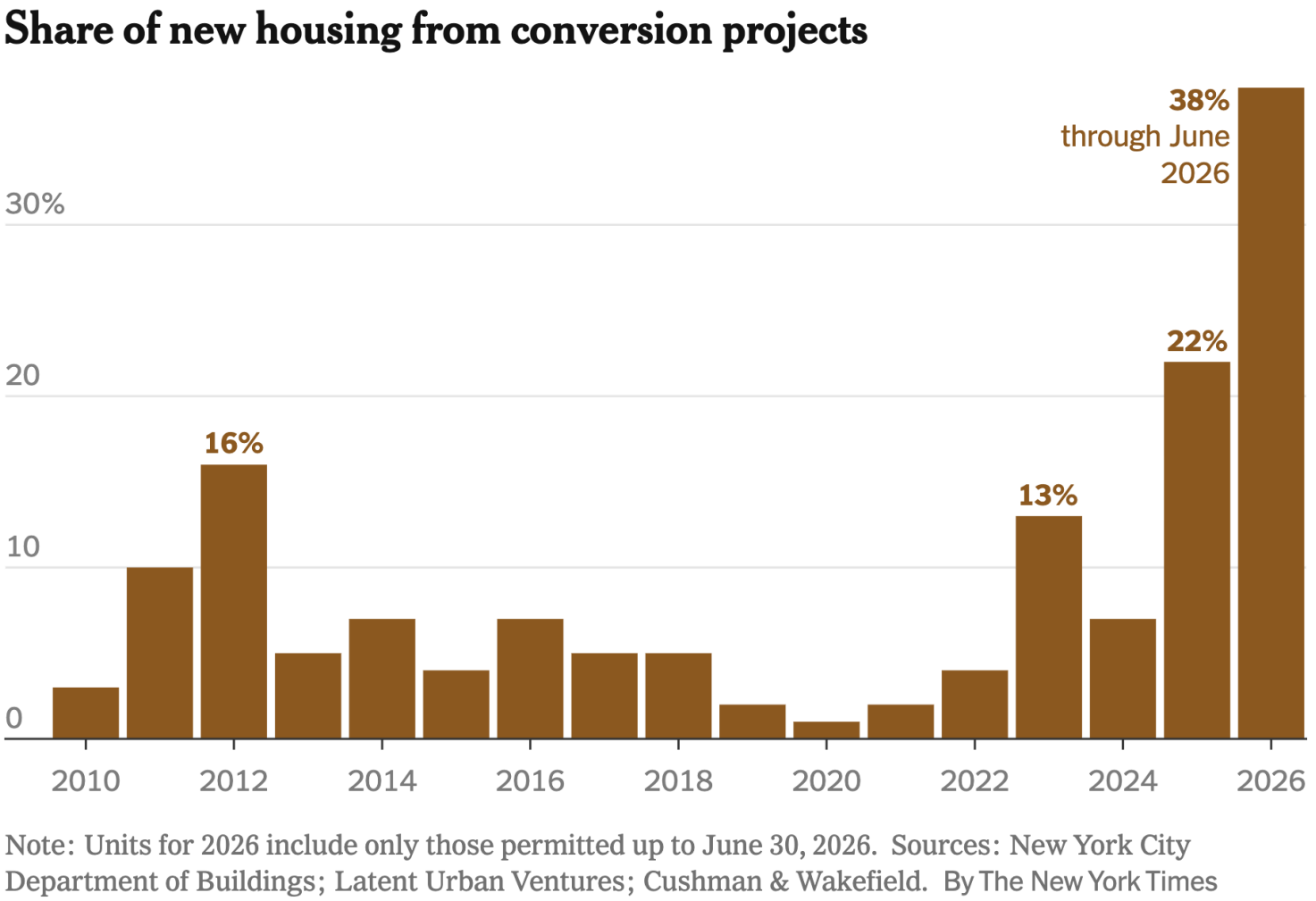
Digital Visualization

Brainstorm

What are the benefits of digital visualization?

Are there any downsides? If so, what?

Example



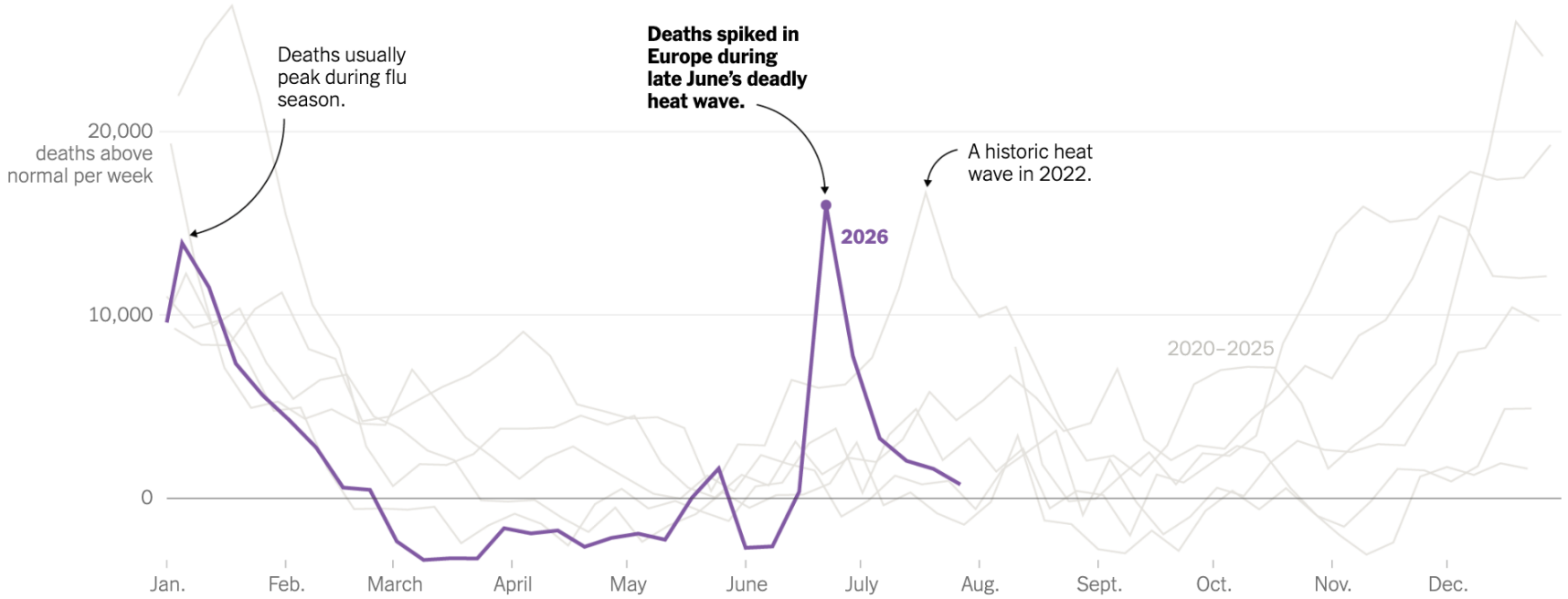
https://www.nytimes.com/2026/08/30/nyregion/nyc-conversions-analysis-pfizer.html

Example

In One Week During Europe's Record Heat Wave, Deaths Increased by 16,000

By Josh Holder Aug. 15, 2026

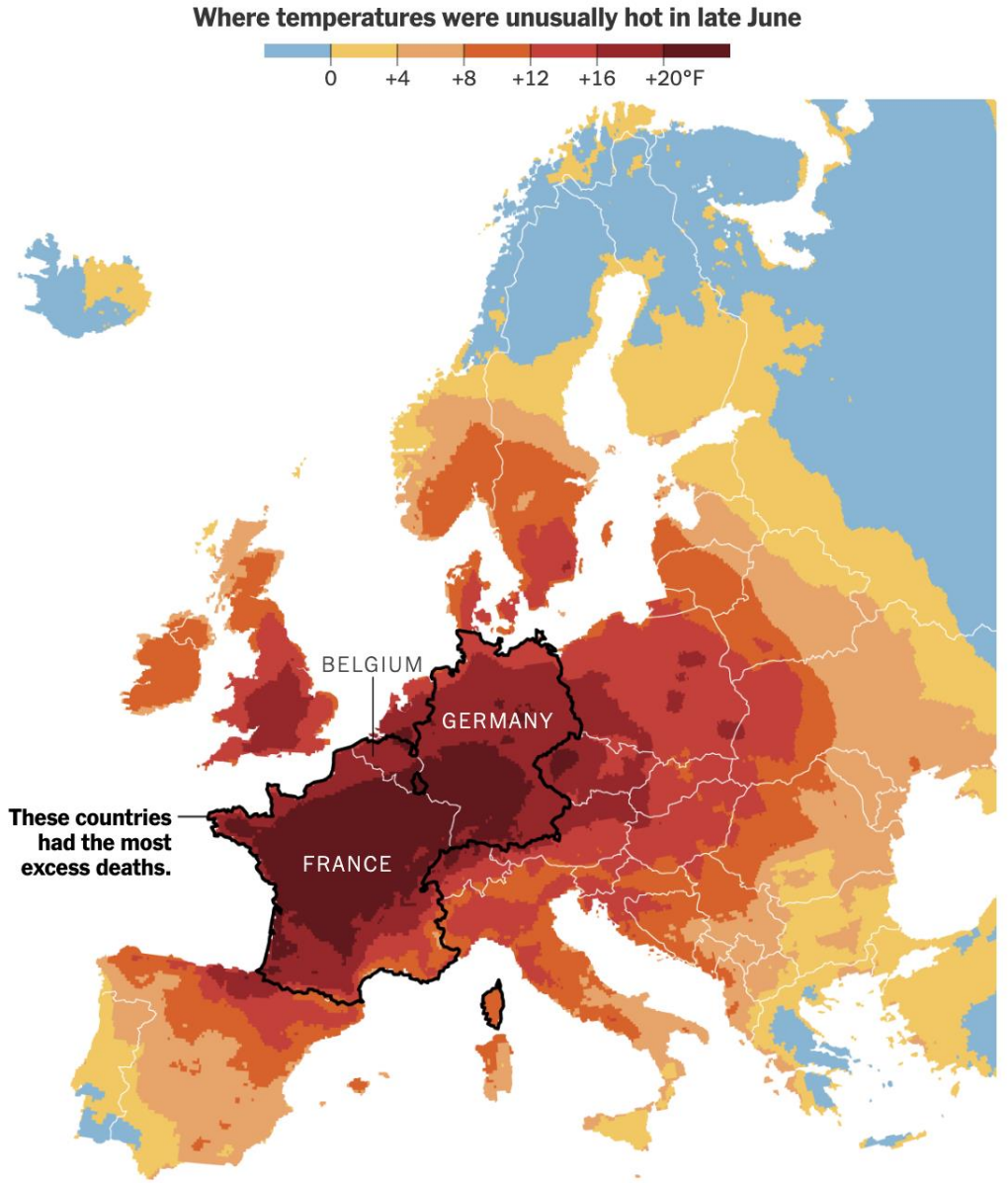
Share full article 169



Note: Includes 23 countries that report data to the European Mortality Monitoring Network. The New York Times

https://www.nytimes.com/interactive/2026/08/15/world/europe/europe-heat-wave-deaths-june-2026.html

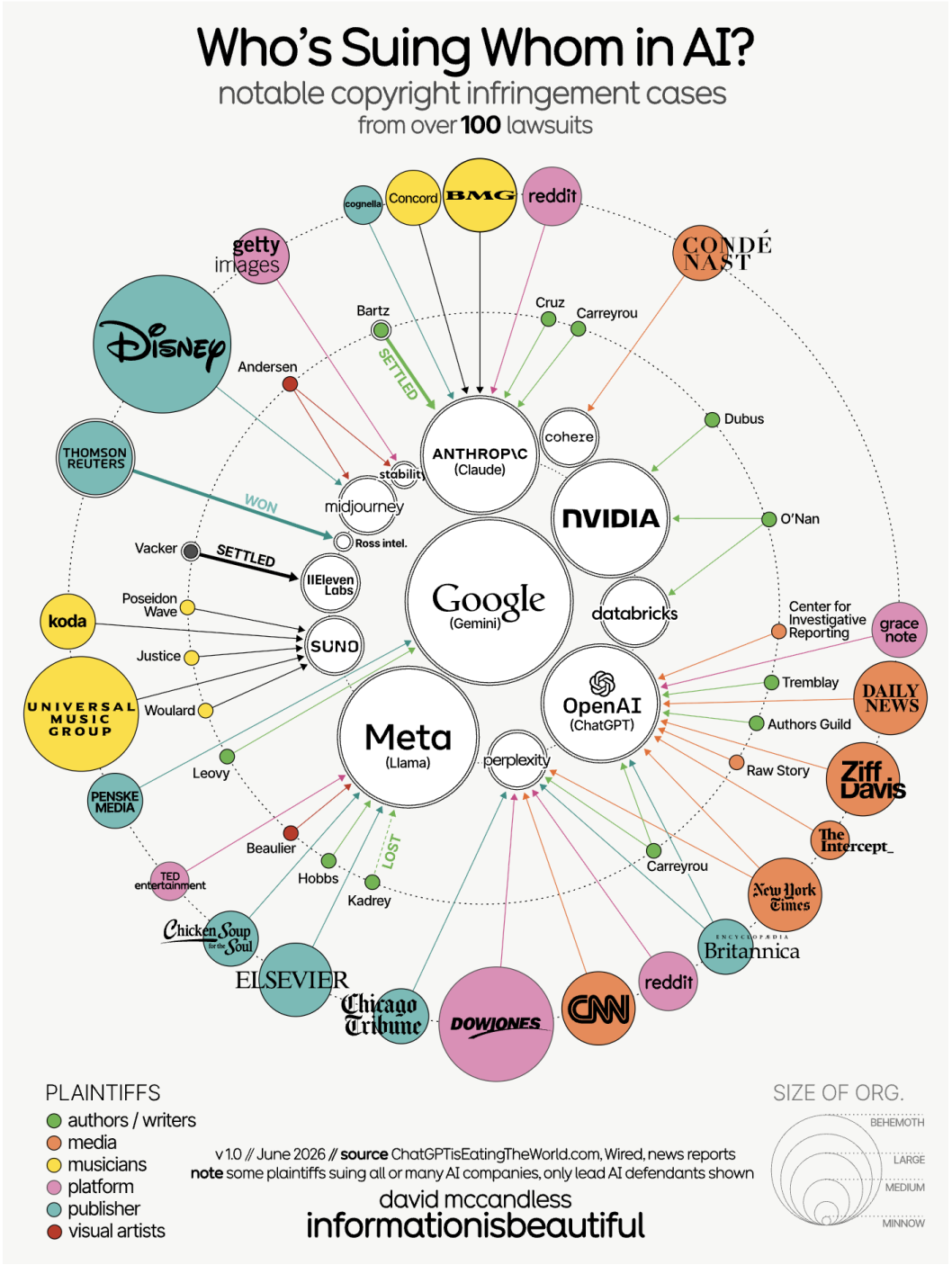
Example



Note: Map shows how much hotter average daily high temperatures were between June 22–28 compared with the average for those same dates in 1991–2020. Source: New York Times analysis of Copernicus ERA5-Land data. The New York Times

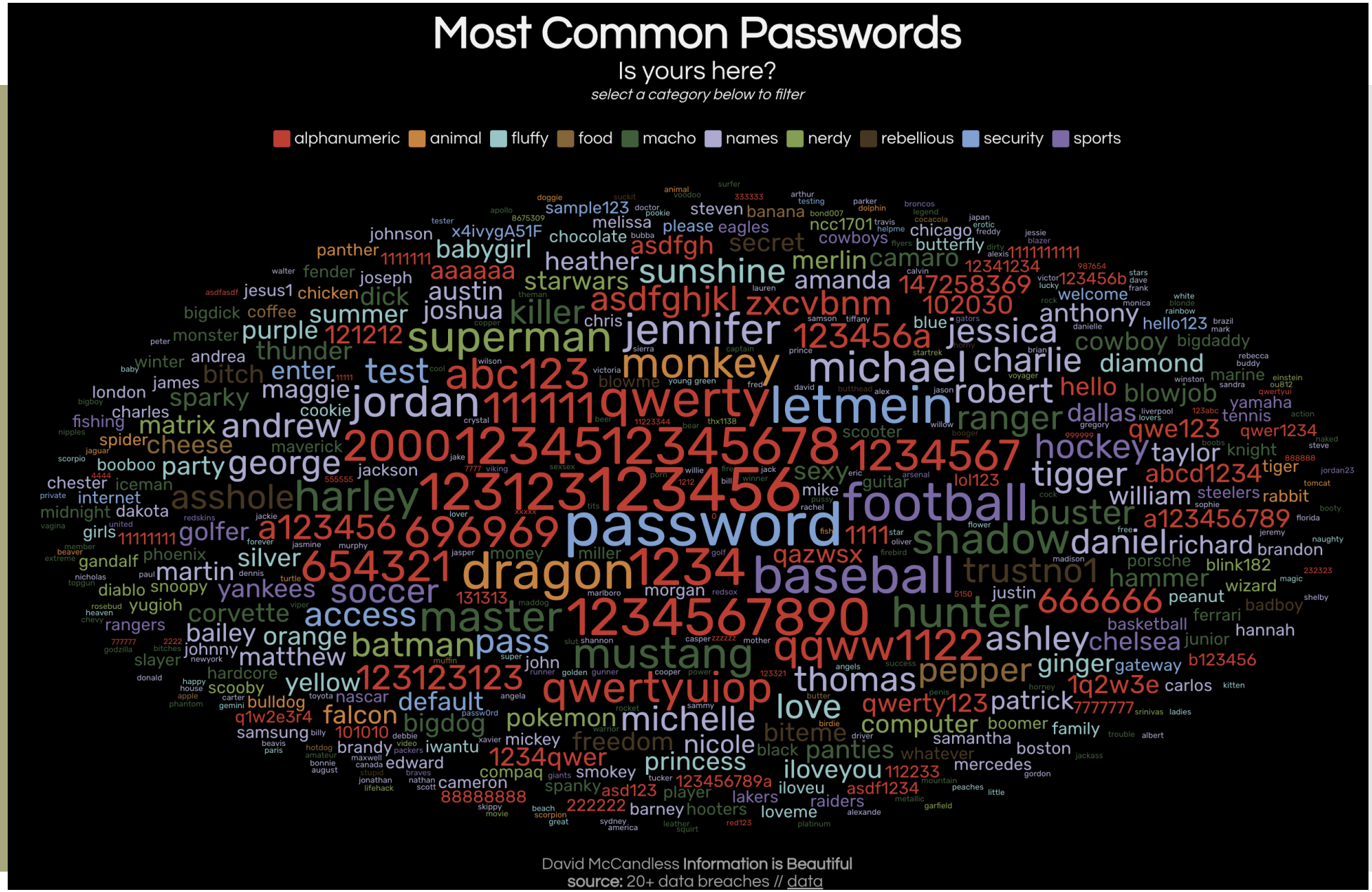
https://www.nytimes.com/interactive/2026/08/15/world/europe/europe-heat-wave-deaths-june-2026.html

Example



https://informationisbeautiful.net/visualizations/the-rise-of-generative-ai-large-language-models-llms-like-chatgpt/#suing

Example



https://informationisbeautiful.net/visualizations/top-500-passwords-visualized/

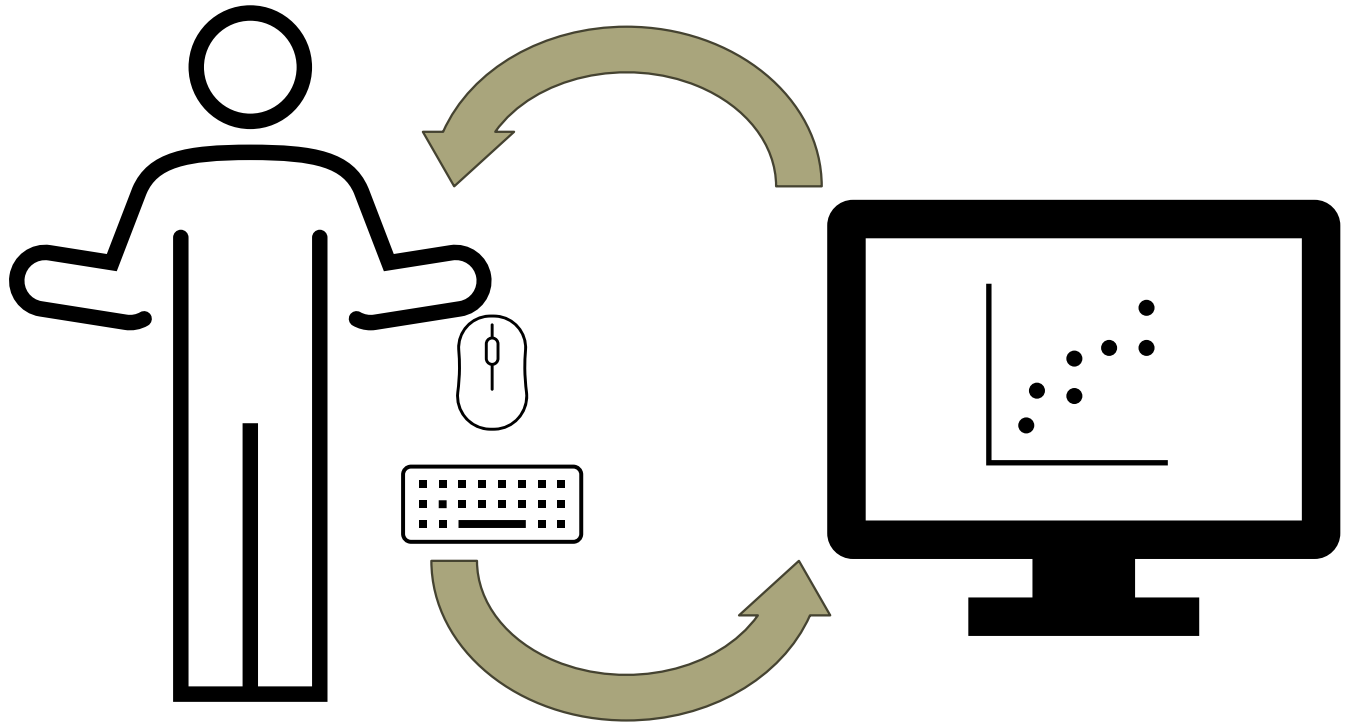
Brainstorm

Choose one of the five previous examples.

What do you think the purpose of the visualization is? In other words, what did the author hope the reader would get from it?

Rewind

Humans and machines have **complimentary strengths**



Interaction

Brainstorm

What is *interaction for visualization*?

Interaction (def.)

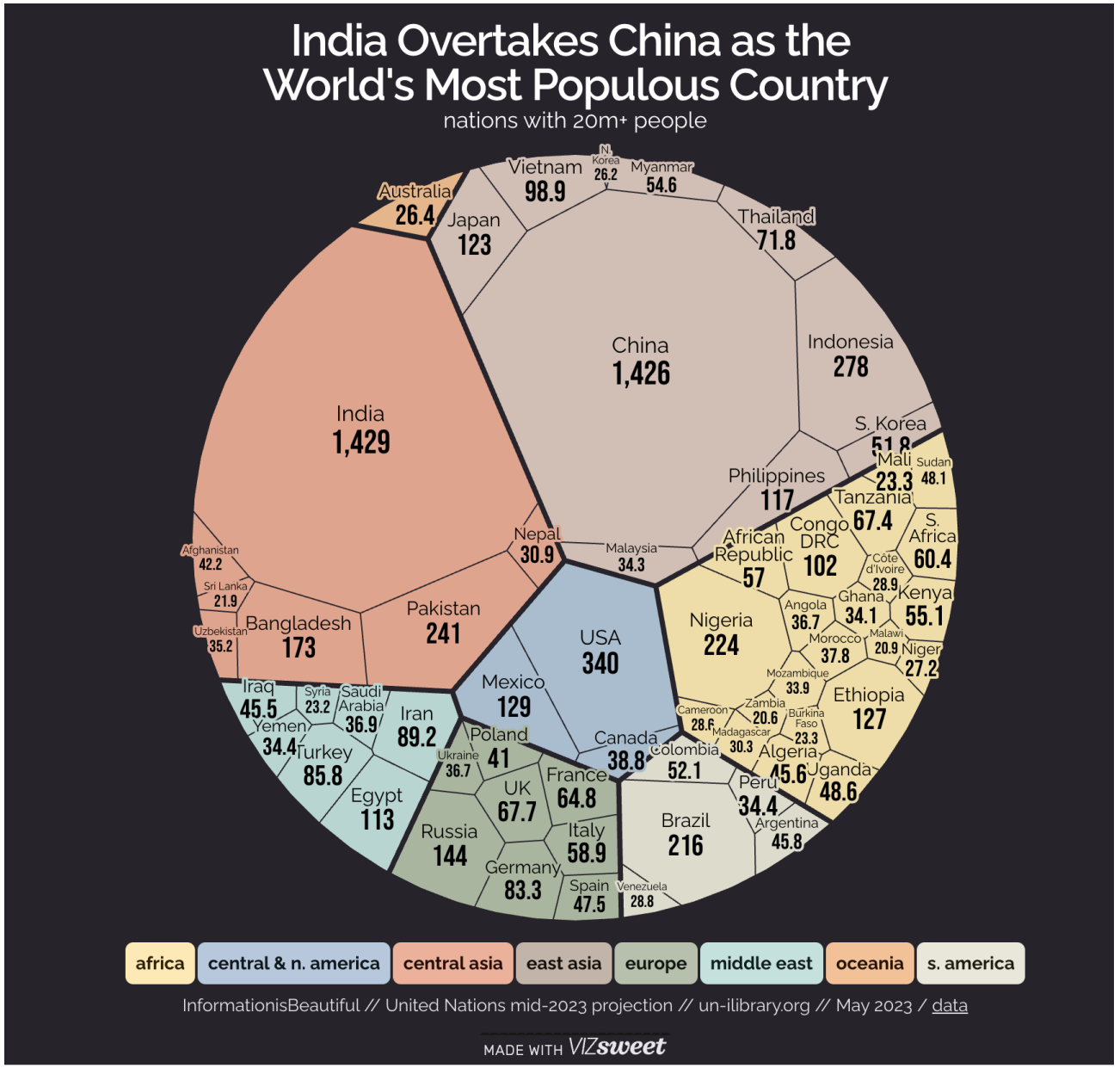
"Interaction for visualization is the interplay between a person and a data interface involving a data-related intent, at least one action from the person and an interface reaction that is perceived as such."

Mandatory Components

- Interplay
- Person
- Data Interface
- Action
- Action-Reaction
- Reaction Perceived as Such
- Data-Related Intent

What is Interaction for Data Visualization? Evanthia Dimara, Charles Perin. 2020

Examples



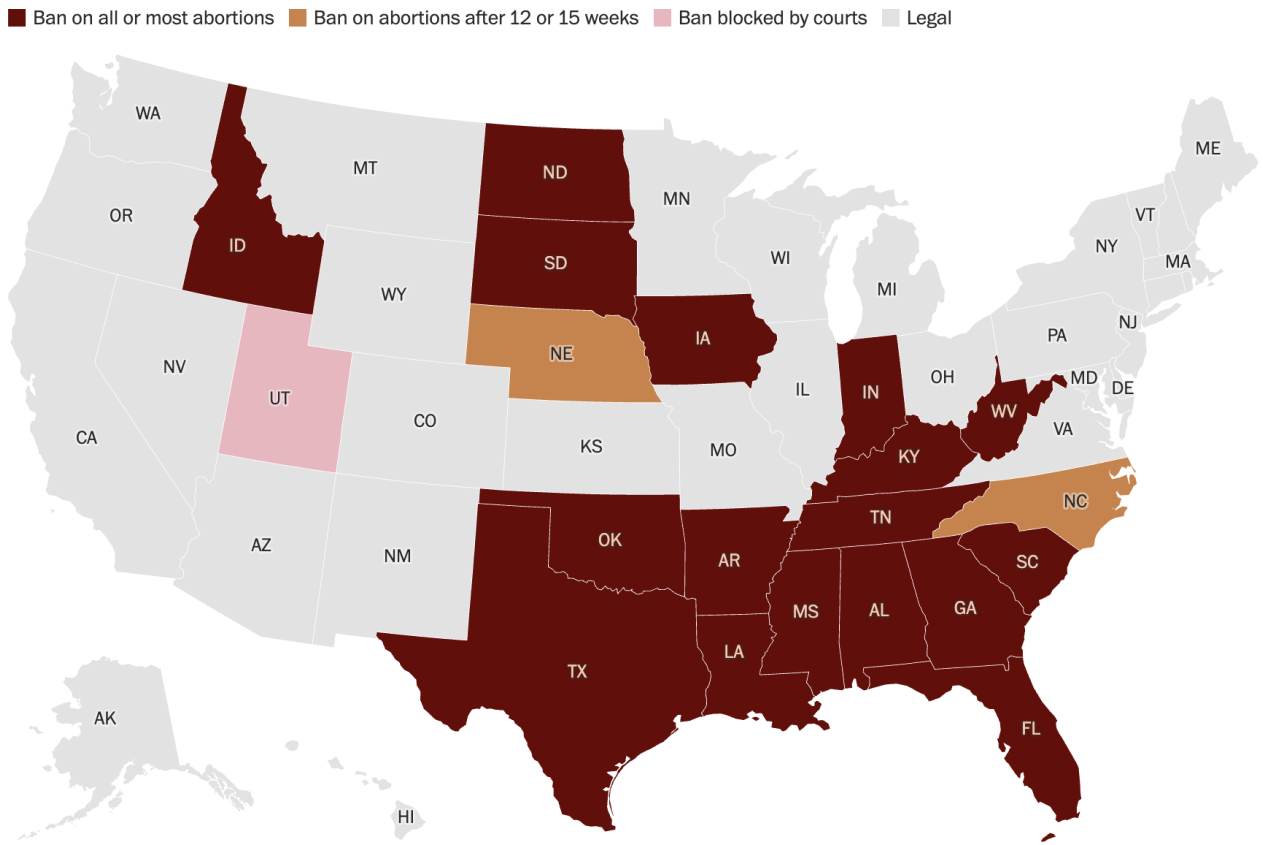
https://informationisbeautiful.net/visualizations/world-population/

Examples

States where abortion is legal, banned or under threat

Updated January 6, 2026 | Published June 24, 2022

Summary 5,427



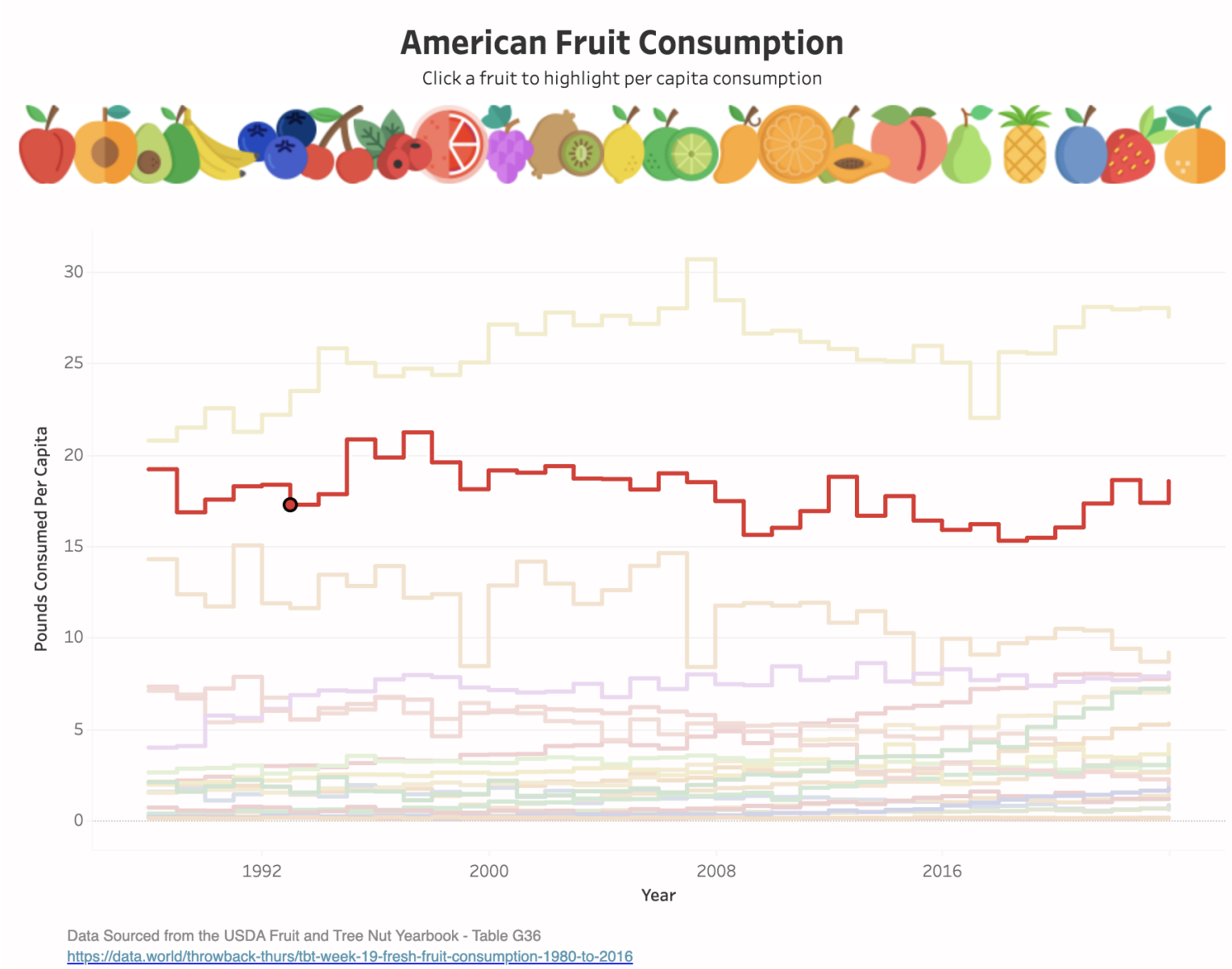
https://www.washingtonpost.com/politics/2022/06/24/abortion-state-laws-criminalization-roe/

Examples



https://public.tableau.com/shared/CC3ZN8YCZ?:display_count=n&:origin=viz_share_link

Examples



https://public.tableau.com/shared/PQ62Y92KP?:display_count=n&:origin=viz_share_link

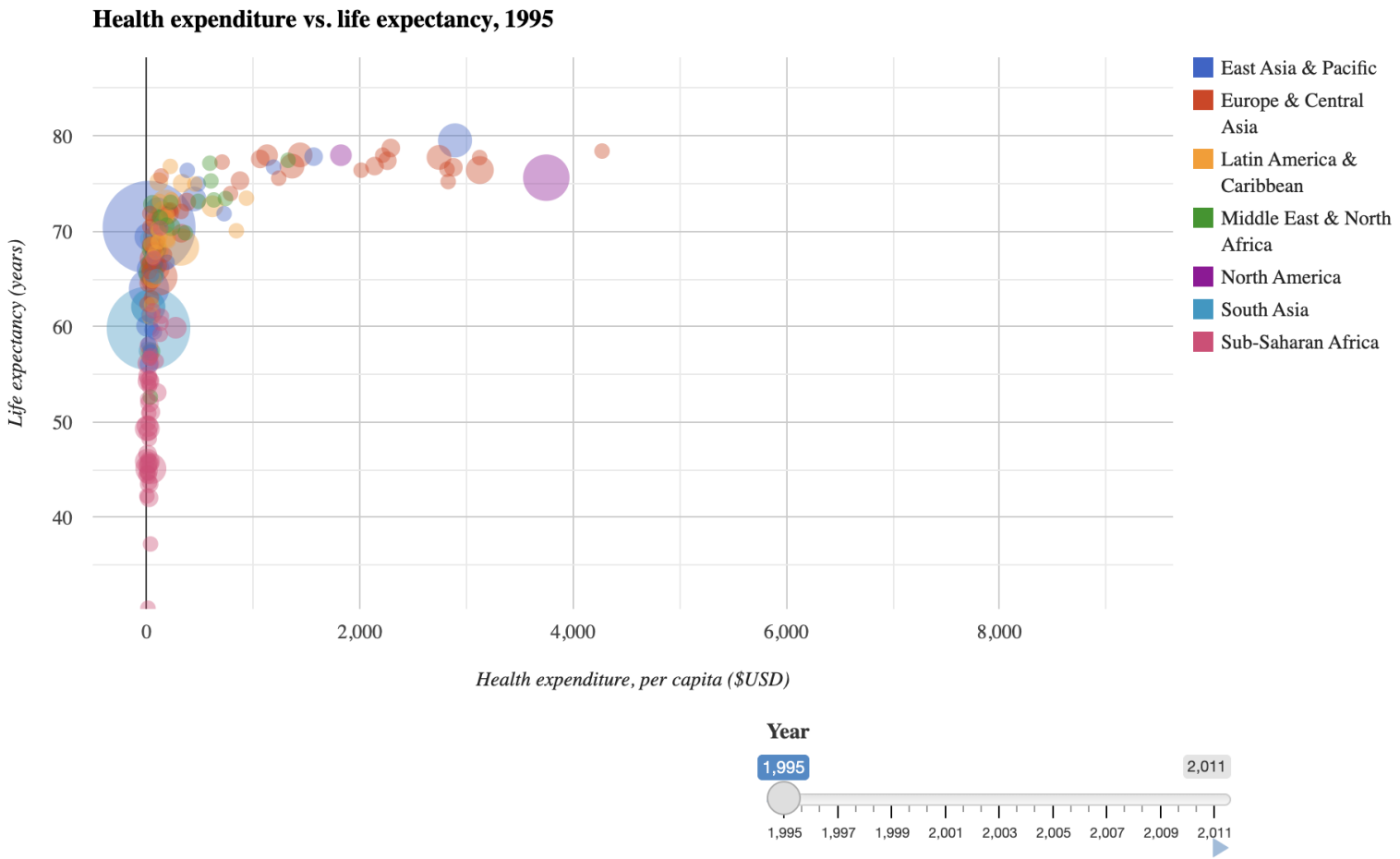
Examples



https://observablehq.com/@d3/brushable-scatterplot-matrix

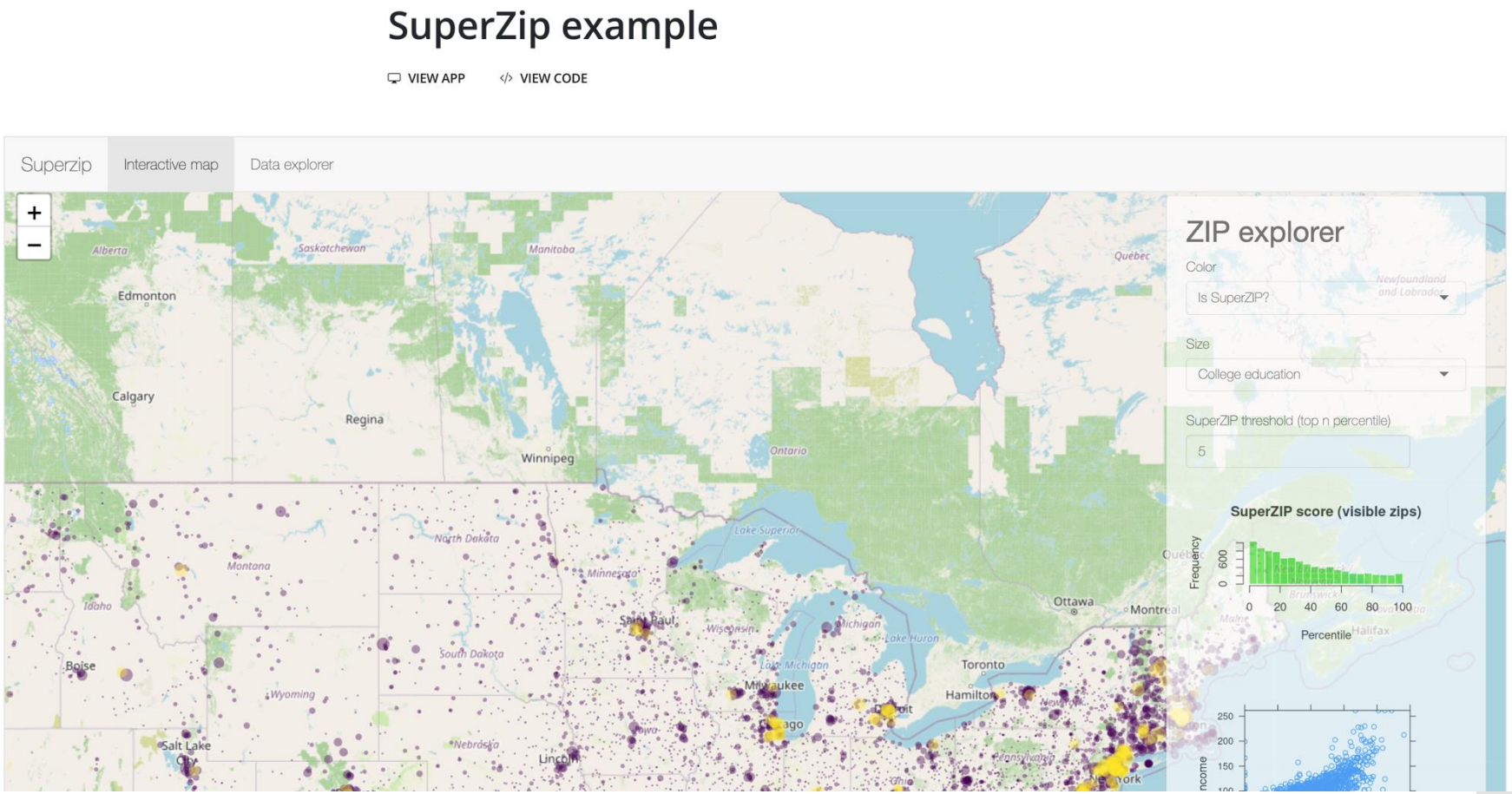
Examples

Google Charts demo



https://gallery.shinyapps.io/182-google-charts/

Examples



https://shiny.posit.co/r/gallery/interactive-visualizations/superzip-example/

Brainstorm

What are two benefits of interaction?

What are two drawbacks of interaction?

Interaction Benefits

- Necessary for handling large amounts of data (more than can fit on a screen)
- Means of amplifying cognition (distributed cognition)
  - Through interactive manipulation "knowledge is constructed, tested, refined, and shared"
- May increase engagement
- May increase deep learning and learning transfer

What is Interaction for Data Visualization? Evanthia Dimara, Charles Perin. 2020
Qiyu Zhi, Alvitta Ottley, and Ronald Metoyer. 2019. Linking and Layout: Exploring the Integration of Text and Visualization in Storytelling. , 675–685 pages.
Chris Evans and Nicola J Gibbons. 2007. The interactivity effect in multimedia learning. Computers & Education 49, 4 (2007), 1147–1160.
Pei-Yu Wang, Brandon K Vaughn, and Min Liu. 2011. The impact of animation interactivity on novices' learning of introductory statistics. Computers & Education 56, 1 (2011), 300–311.

Interaction Drawbacks

- Experience and research suggest readers do not choose to interact
- Studies on adding interaction to visualizations that could otherwise be static show no significant gains

Why We Are Doing Fewer Interactives (Archie Tse, The New York Times): https://github.com/archietse/malofiej-2016/blob/master/tse-malofiej-2016-slides.pdf
Ab Mosca, Alvitta Ottley, and Remco Chang. 2021. Does Interaction Improve Bayesian Reasoning with Visualization? In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 609, 1–14. https://doi.org/10.1145/3411764.3445176

Brainstorm

List all the ways a reader could interact with a visualization.



One conceptualization

High level: **between human and problem space**

- a cognitive act *enabled* by the tool
- does not need to take place exclusively within them
- might be distributed across multiple tools

Low level: **between human and interface**

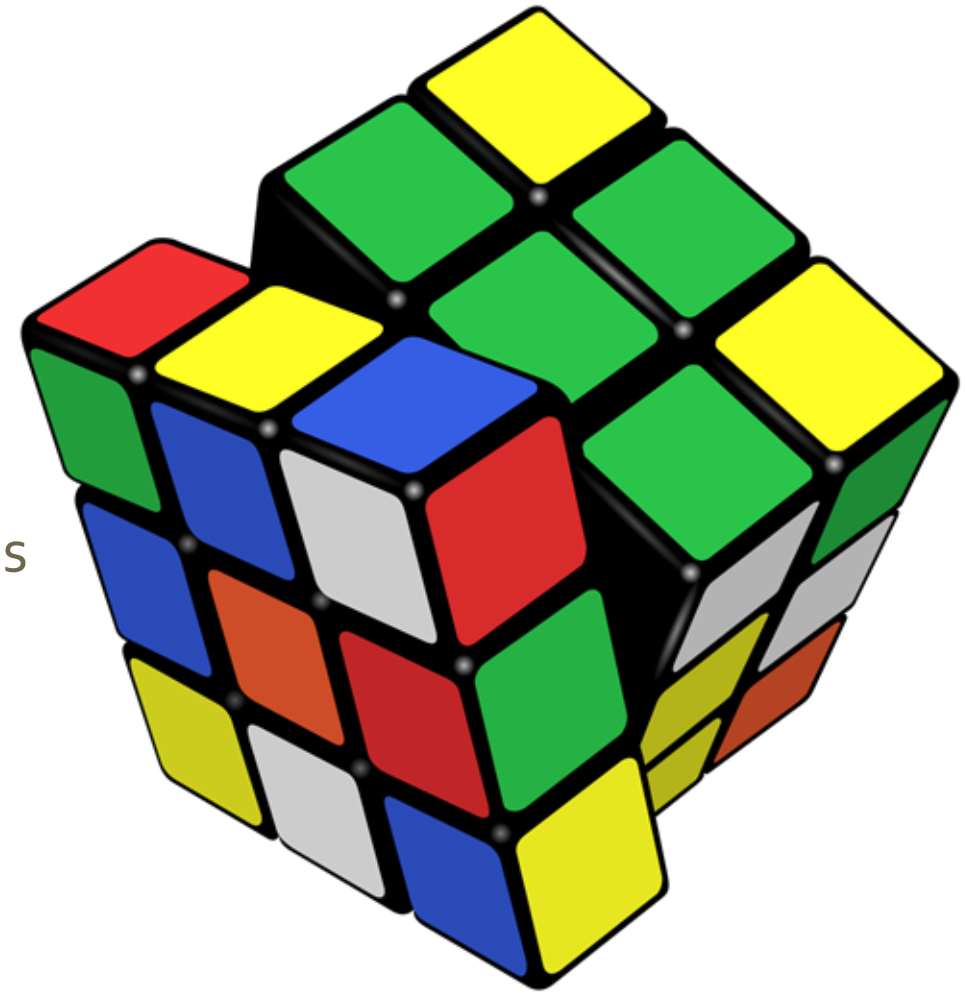
- the set of operations available
- the relationship between the human and the visualization

Pike, W. A., Stasko, J., Chang, R., & O'Connell, T. A. (2009). The Science of Interaction. *Information Visualization*, *8*(4), 263-274. https://doi.org/10.1057/ivs.2009.22

Example: Rubik's Cube

What **high-level** interactions are there?

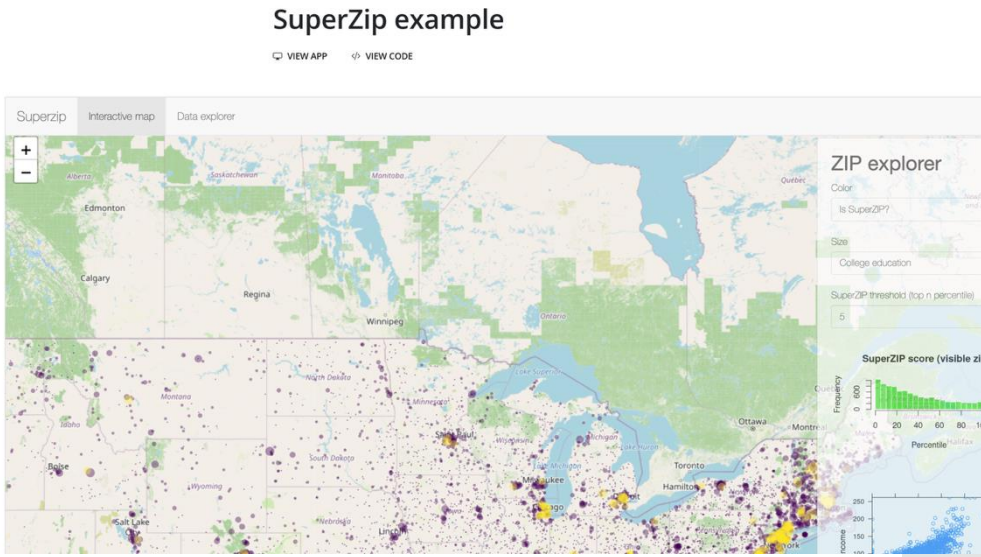
What **low-level** interactions could there be?



Example: SuperZip

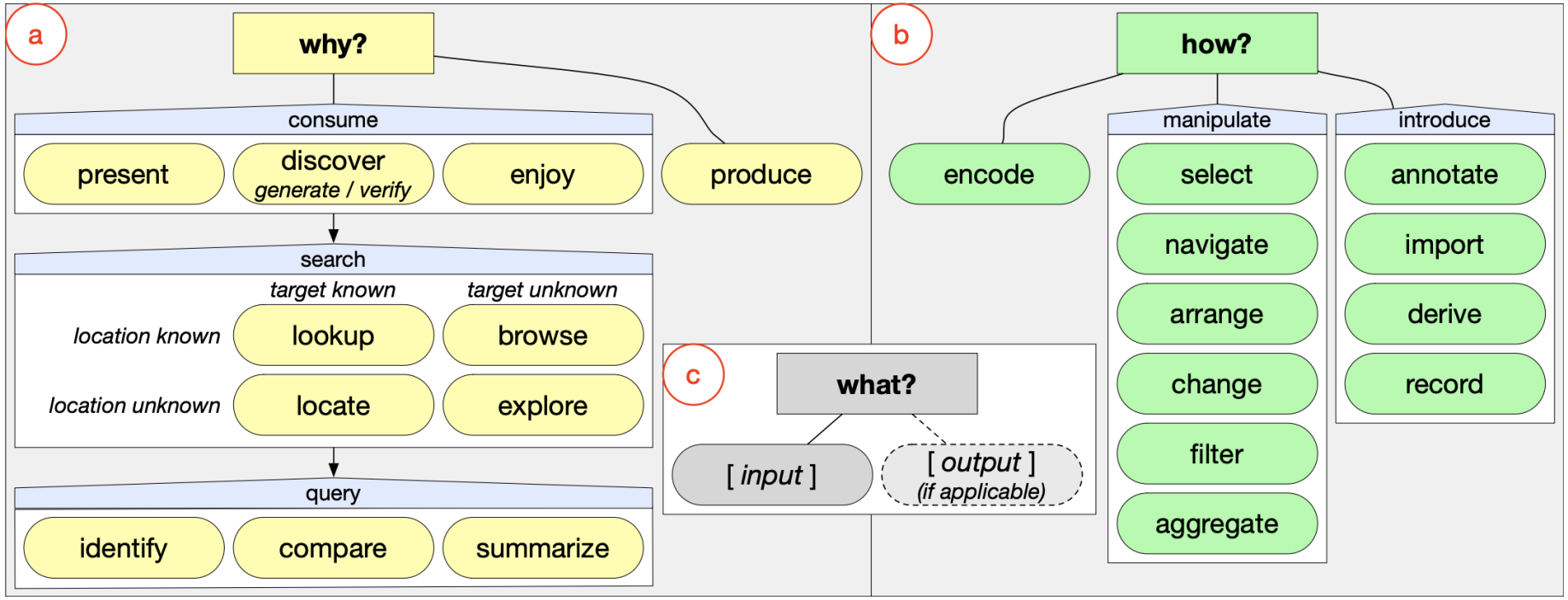
What **high-level** interactions are there?

What **low-level** interactions are there?



https://shiny.posit.co/r/gallery/interactive-visualizations/superzip-example/

A different conceptualization



M. Brehmer and T. Munzner, "A Multi-Level Typology of Abstract Visualization Tasks," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 19, no. 12, pp. 2376-2385, Dec. 2013, doi: 10.1109/TVCG.2013.124.

A third conceptualization

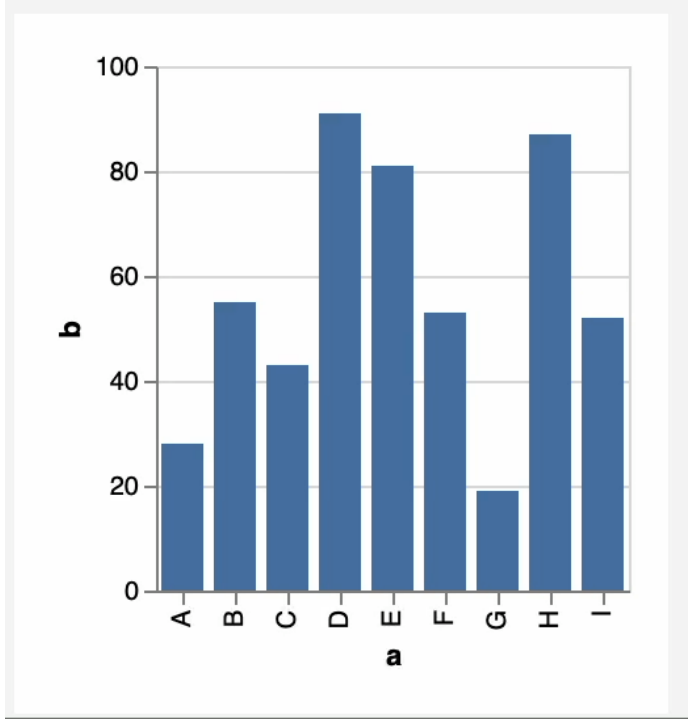
Seven general categories of interaction (by user intent):

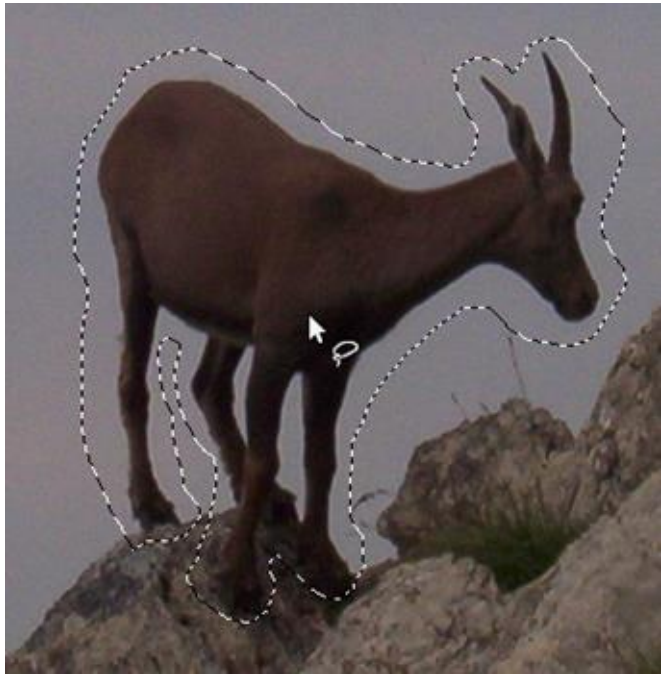
1. Select
2. Explore
3. Reconfigure
4. Encode
5. Abstract/Elaborate
6. Filter
7. Connect

Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Select
Mark something as interesting: direct



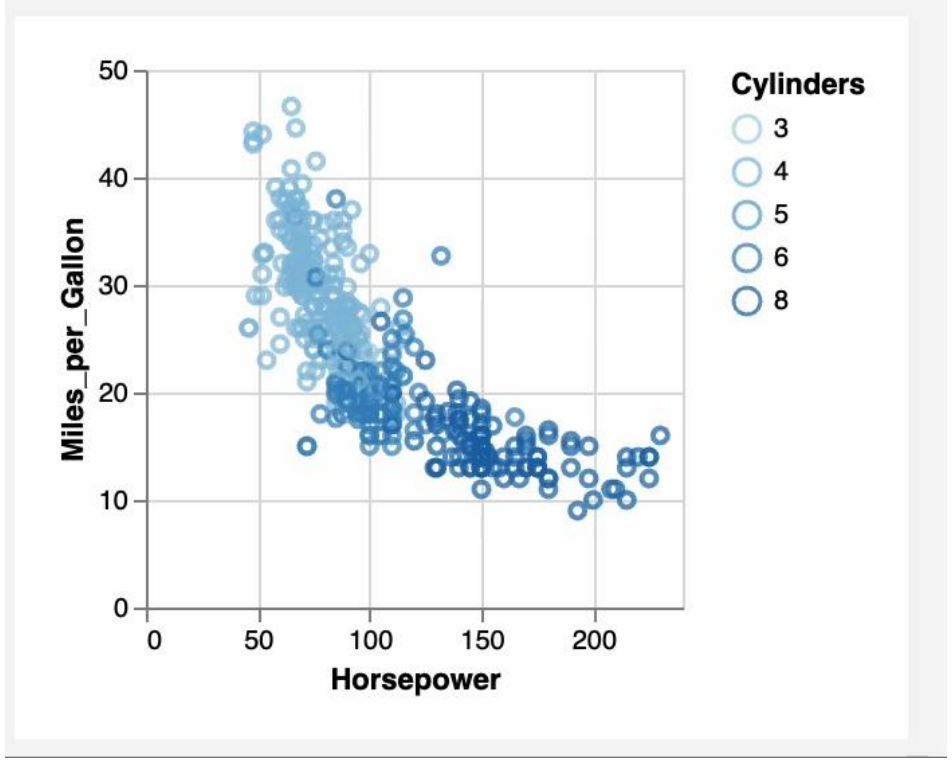


Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Select

Mark something as interesting: indirect



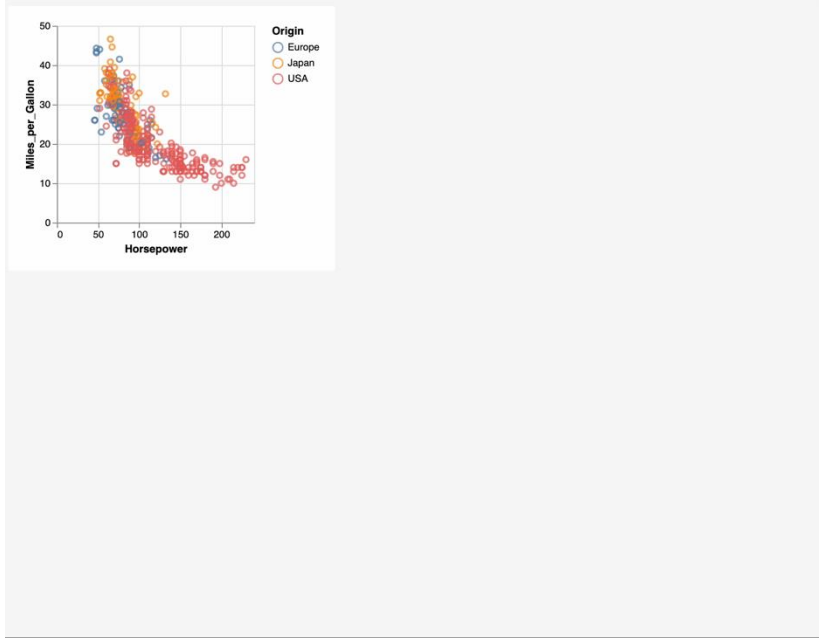
Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Explore

Show me something else

- Scroll bars
- Panning
- Direct-Walk (e.g. hyperlink traversal)

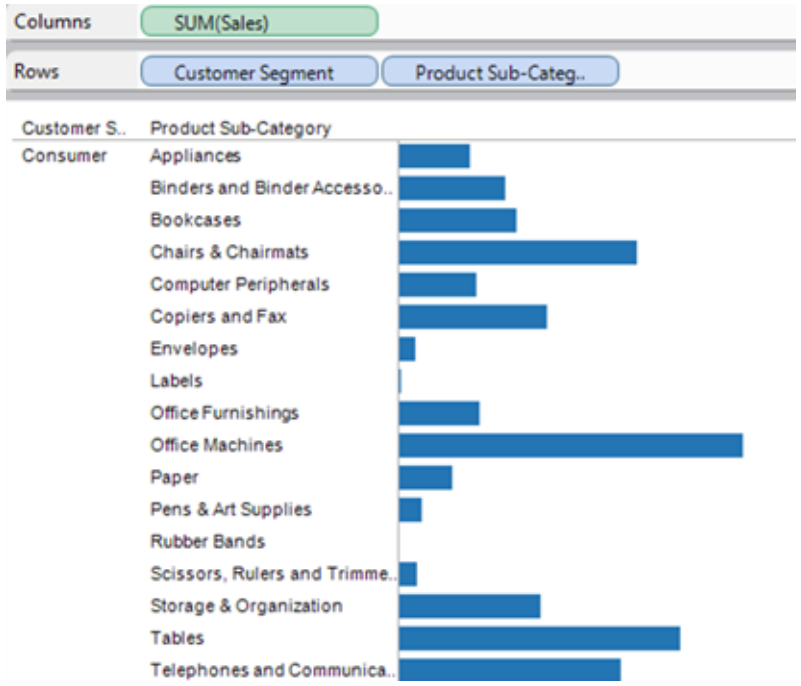


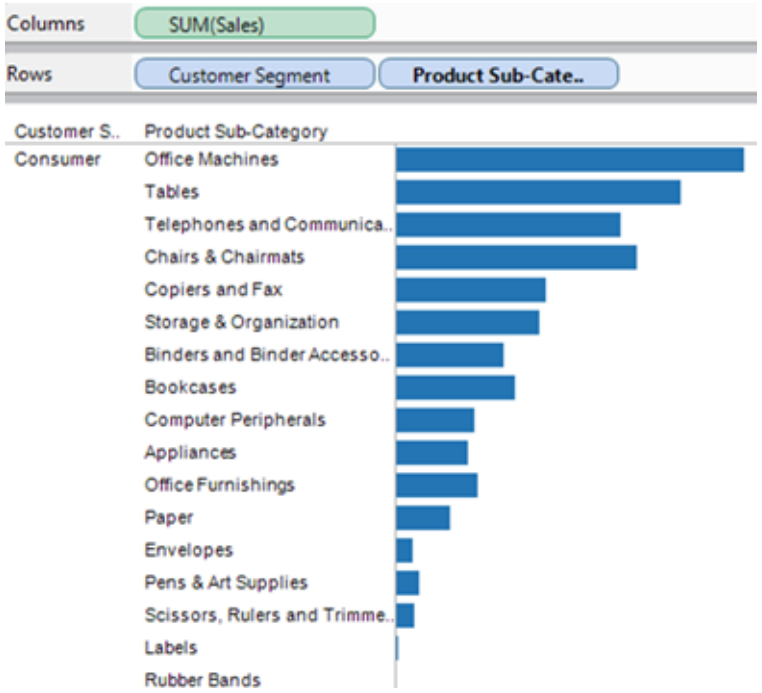
Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Reconfigure

Show me a different arrangement: sorting



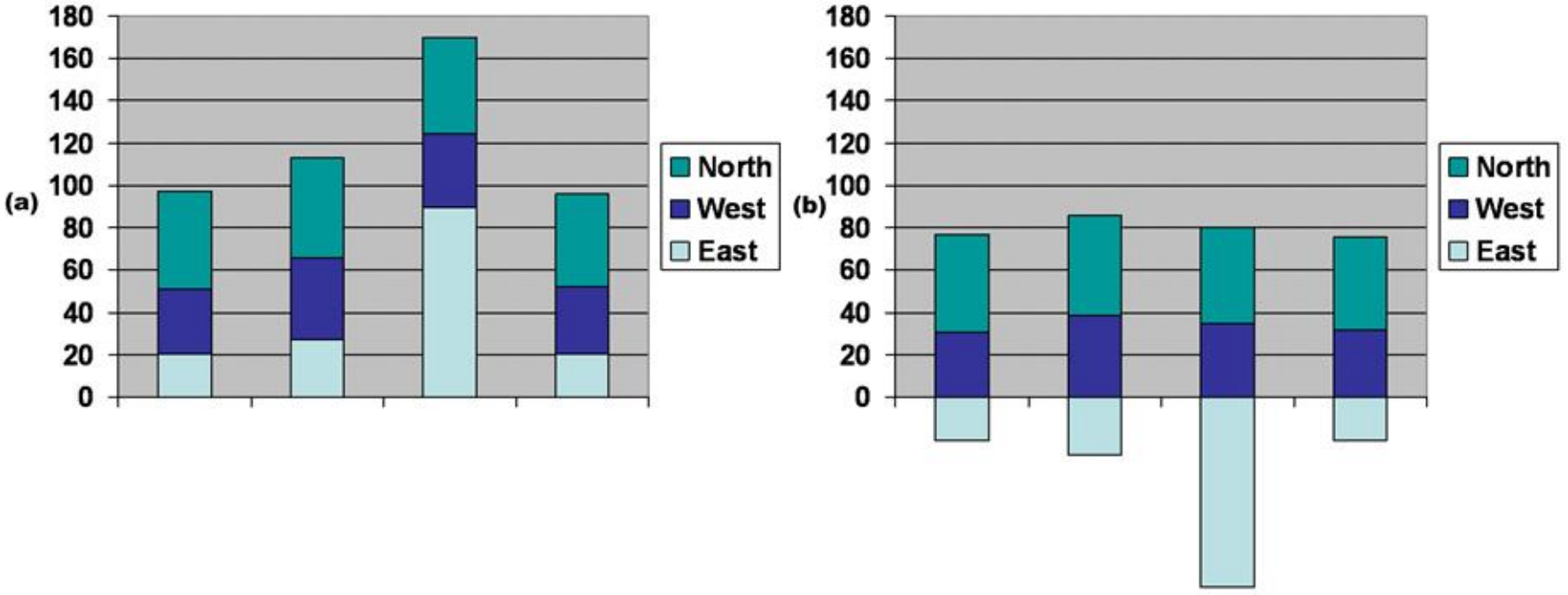


Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Reconfigure

Show me a different arrangement: baseline adjustment

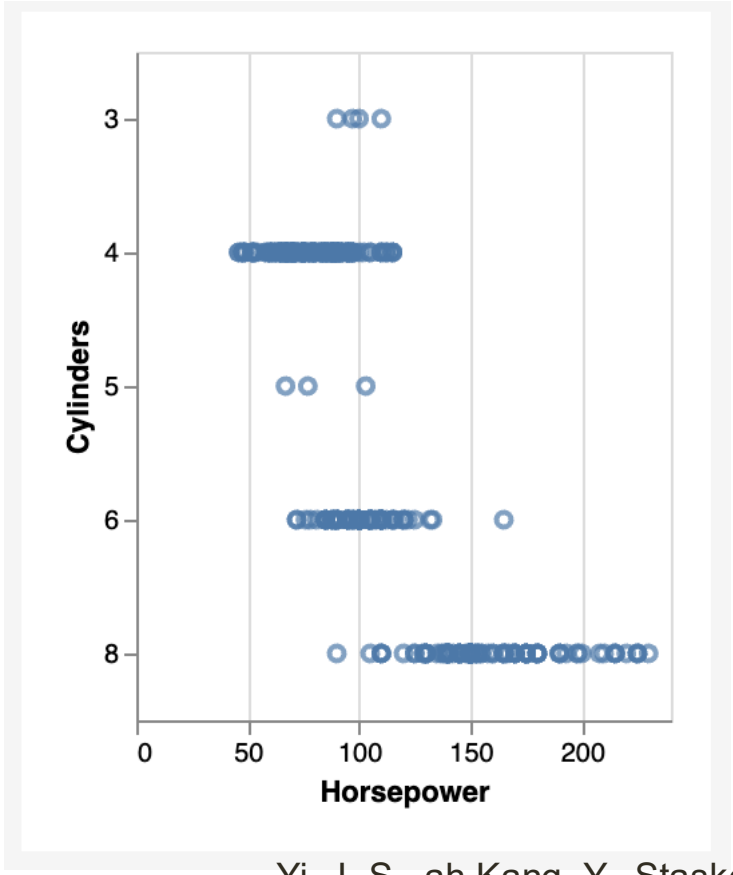


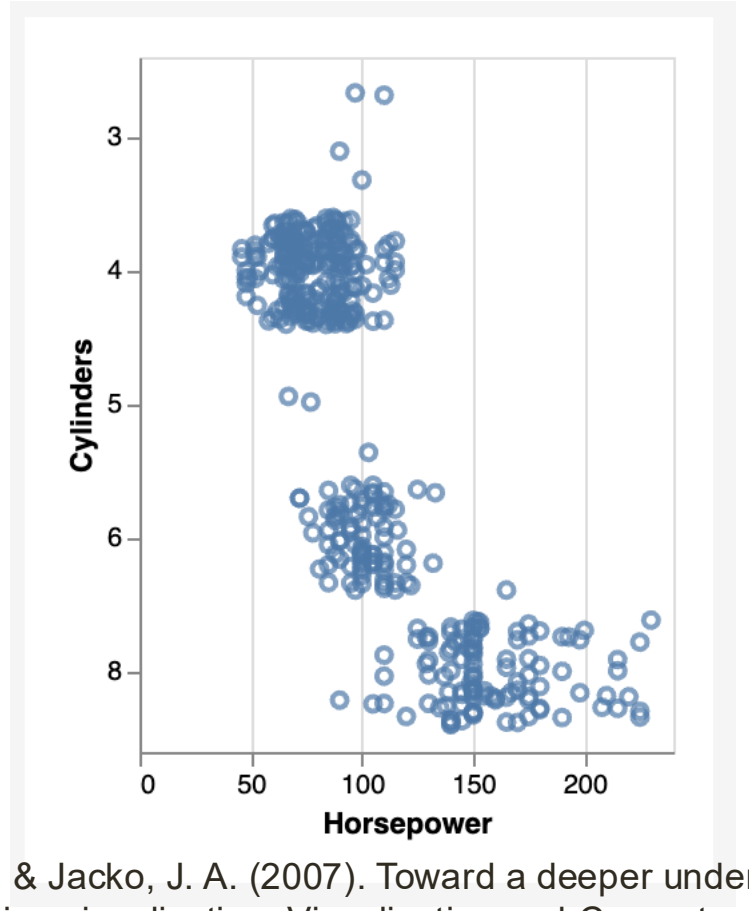
Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Reconfigure

Show me a different arrangement: reduce occlusion (jitter)





Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Encode

Show me a different representation: visualization type, color, size, orientation, etc.

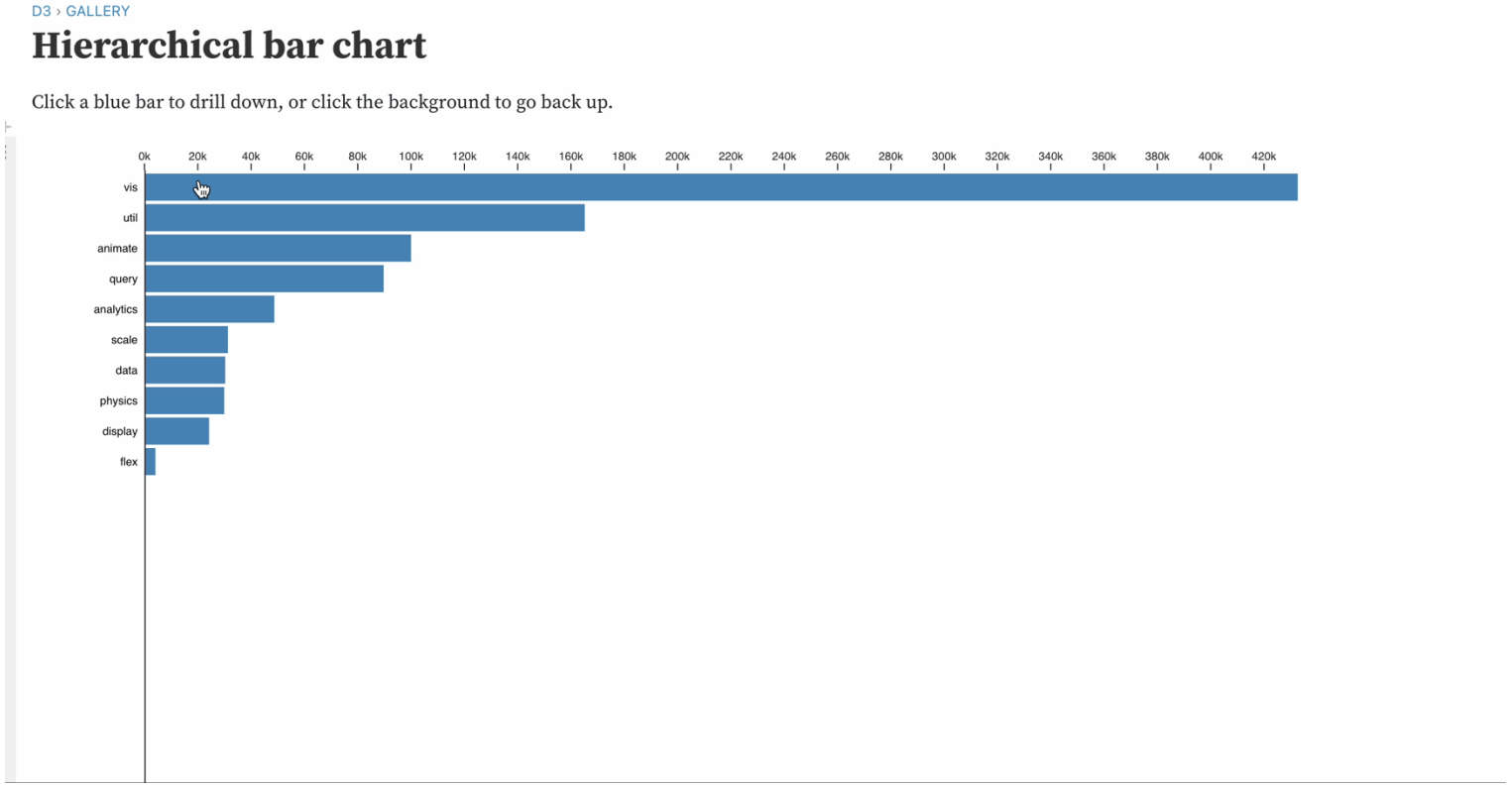


Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Abstract/Elaborate

Show me more or less detail: drill up/down

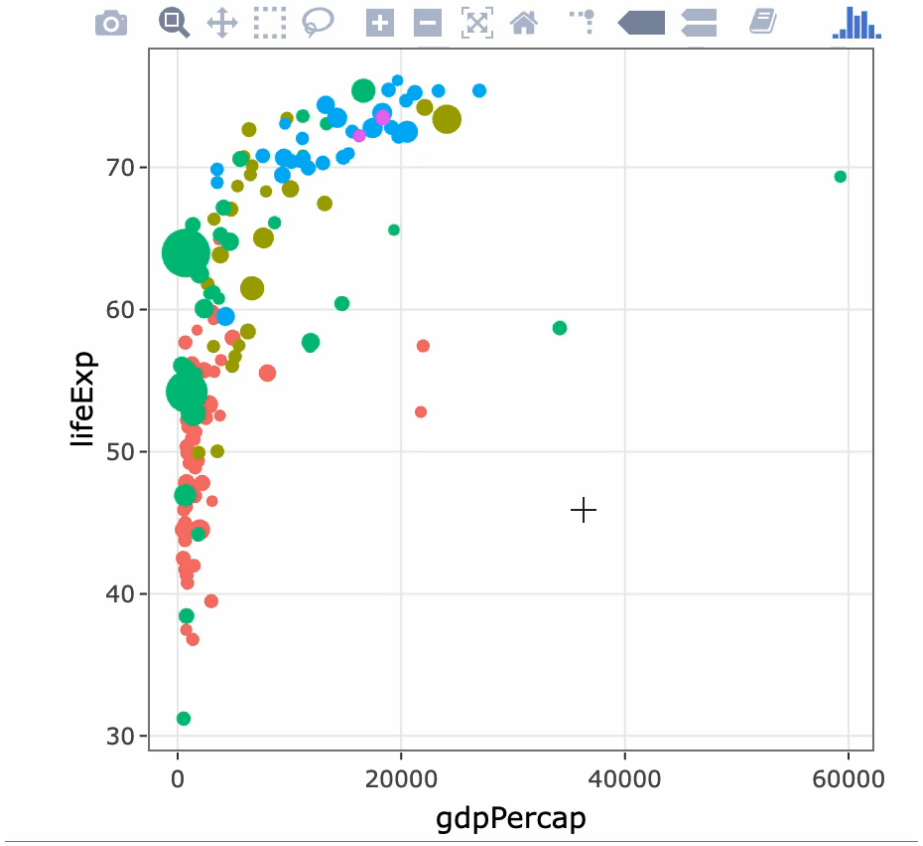


Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Abstract/Elaborate

Show me more or less detail: zooming

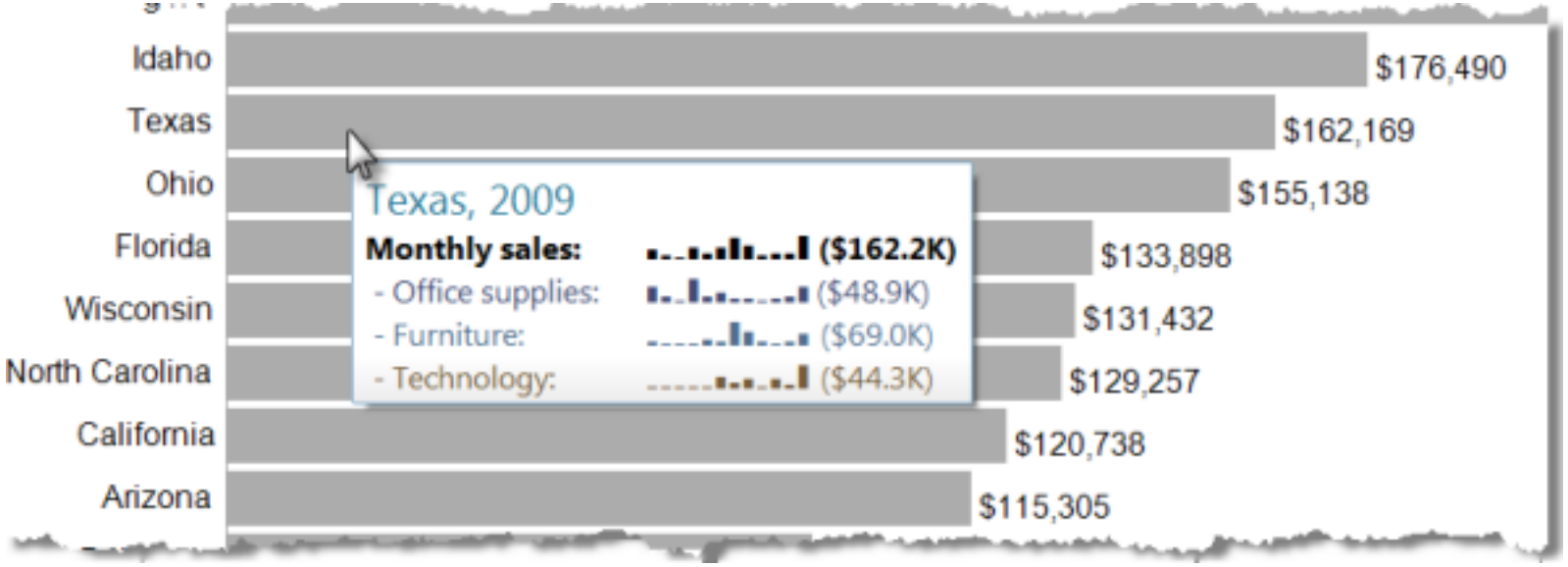


Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Abstract/Elaborate

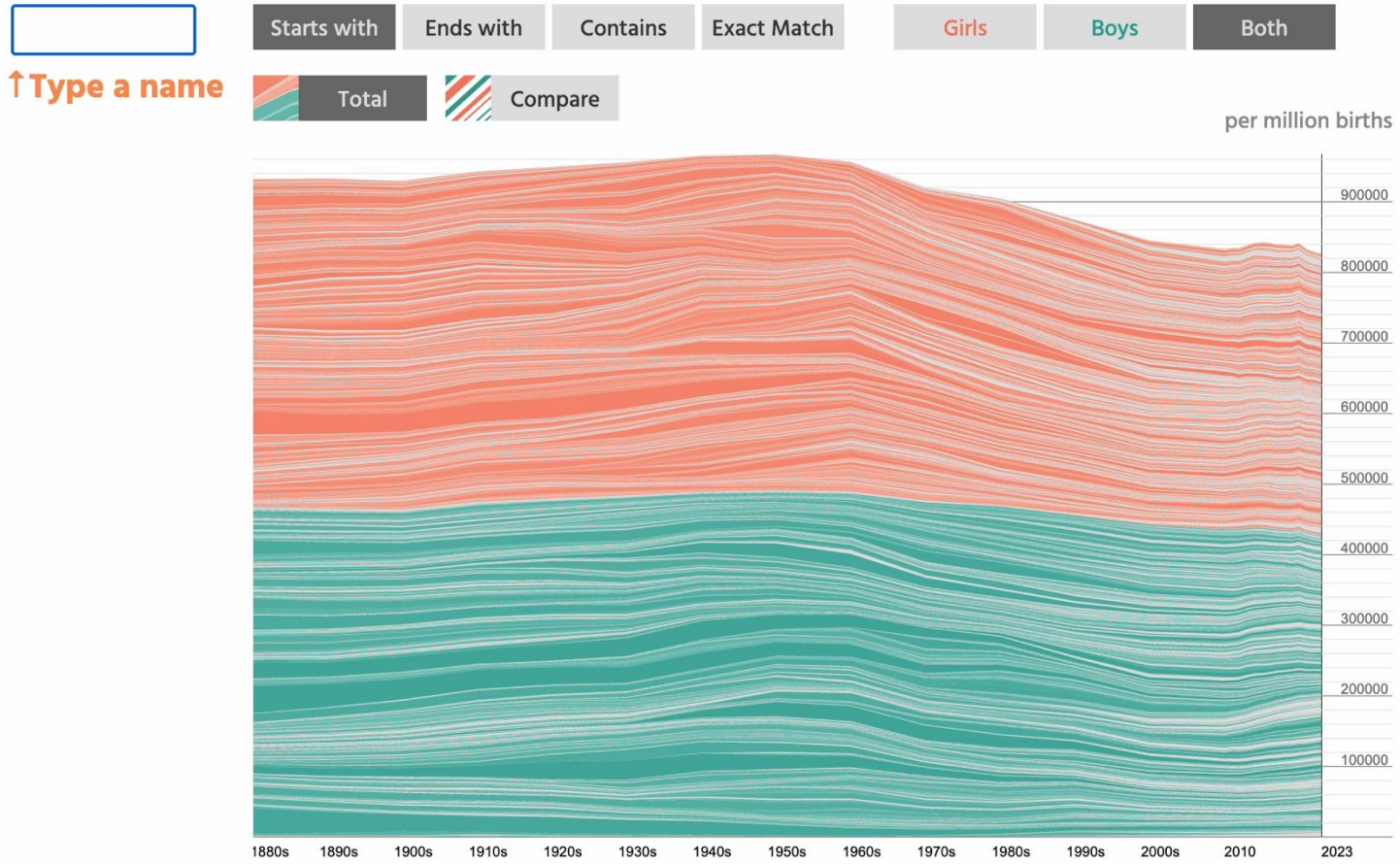
Show me more or less detail: tooltips



Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

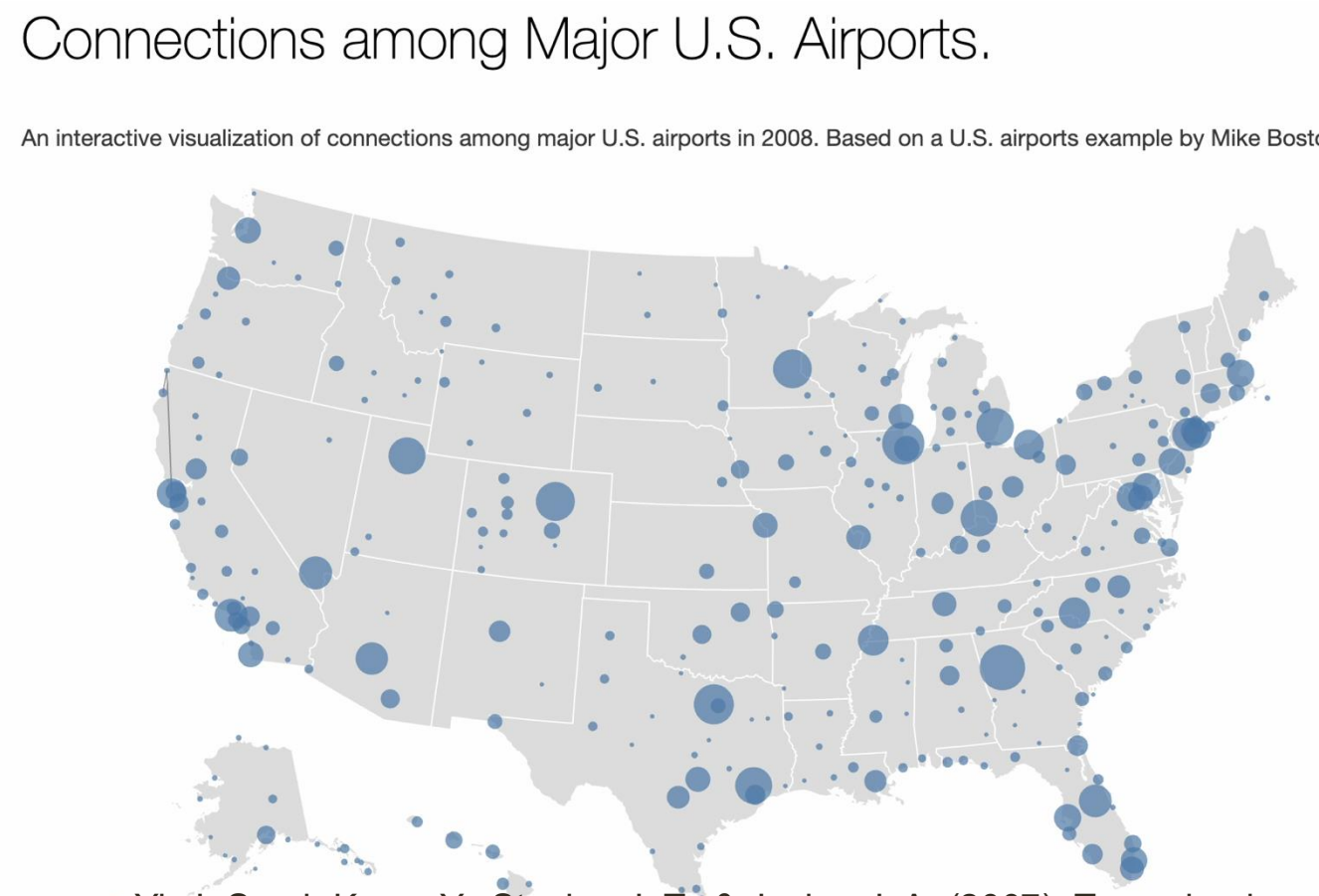
Filter
Show me something conditionally



Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

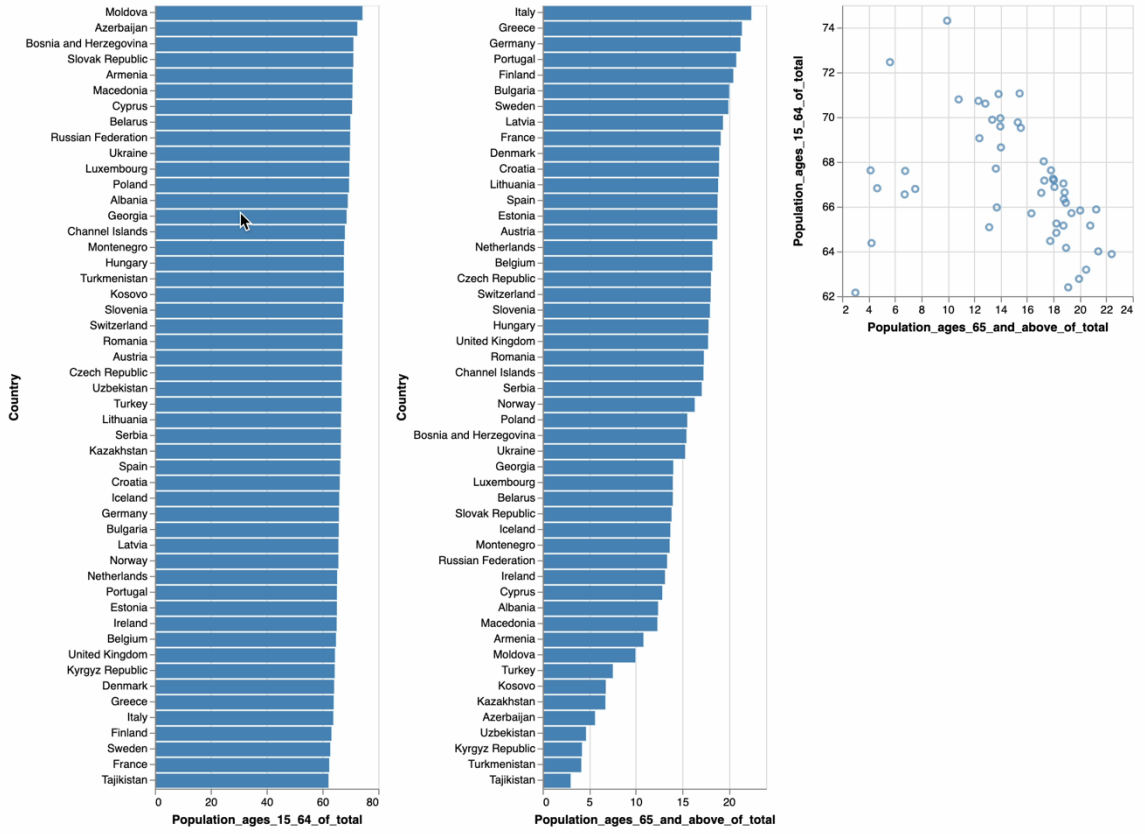
Connect
Show me related items: build-out



Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

A third conceptualization

Connect
Show me related items: coordinated multiple views (CMV)



Yi, J. S., ah Kang, Y., Stasko, J. T., & Jacko, J. A. (2007). Toward a deeper understanding of the role of interaction in information visualization. Visualization and Computer Graphics, IEEE Transactions on, 13(6), 1224-1231.

Your Turn!

Return to one of the interactive examples from earlier, or spend <= 5 minutes finding a new one.

Use each conceptualization approach (1: high level vs. low level, 2: why & how, 3: seven categories) to categorize the different interactions in the visualization.